Network Intrusion Detection System (NIDS) – Interview Revision Report

Executive Summary

This project focuses on building a Network Intrusion Detection System (NIDS) using unsupervised machine learning techniques for anomaly detection in cybersecurity traffic data. The notebook combines clustering, dimensionality reduction, and anomaly detection approaches to identify suspicious network behavior without depending on predefined attack signatures.

The system uses:

- Principal Component Analysis (PCA) for dimensionality reduction
- MiniBatch K-Means for traffic clustering
- Isolation Forest for anomaly detection
- One-Class SVM for boundary-based anomaly identification
- A Hybrid Ensemble approach for combined anomaly prediction

The dataset contains approximately 286,467 rows and 79 columns after preprocessing. Extensive data cleaning, exploratory data analysis (EDA), feature engineering, scaling, PCA transformation, and visualization were performed before training the models.

The project demonstrates how unsupervised learning can be applied in cybersecurity environments where labeled attack data is either unavailable or incomplete. The final solution identifies abnormal traffic behavior patterns and groups suspicious activities into clusters.

This report serves as a complete interview revision guide explaining:

- Dataset understanding
- EDA insights
- Feature engineering decisions
- PCA interpretation
- Clustering analysis
- Anomaly detection algorithms
- Cybersecurity implications
- Visualization interpretations
- Resume-ready project points
- Interview preparation questions and answers

Abstract

Traditional signature-based intrusion detection systems struggle to identify zero-day attacks and evolving threats. Machine learning-driven anomaly detection provides an adaptive approach capable of recognizing suspicious traffic patterns even when attack signatures are unknown.

This project implements an unsupervised Network Intrusion Detection System (NIDS) using multiple machine learning algorithms. The workflow includes:

1. Data preprocessing and cleaning
2. Feature engineering
3. Feature scaling
4. Principal Component Analysis (PCA)
5. Clustering using MiniBatch K-Means
6. Anomaly detection using Isolation Forest
7. Boundary detection using One-Class SVM
8. Hybrid anomaly scoring and evaluation

The project emphasizes scalability and practical applicability in real-world cybersecurity monitoring systems.

The key objective is to distinguish normal traffic behavior from anomalous or potentially malicious behavior using statistical and geometric properties of the dataset.

The clustering visualizations provide interpretability by helping analysts understand the structure of network traffic. The anomaly detection algorithms further classify suspicious activities based on isolation, density, and distance-based properties.

Although the dataset contains attack labels for evaluation, the learning process itself is unsupervised, making the system more realistic for production-level security operations.

Project Architecture Overview

End-to-End Pipeline

Step 1: Data Loading

The dataset is loaded using Pandas and inspected for:

- Missing values
- Infinite values
- Duplicate records
- Datatype inconsistencies

Step 2: Data Cleaning

The preprocessing pipeline removes:

- Whitespaces
- NaN values
- Infinite values
- Duplicate rows

This ensures stable model behavior.

Step 3: Exploratory Data Analysis

EDA is performed to:

- Understand traffic distribution
- Analyze attack imbalance
- Study correlations
- Identify skewness and variance
- Understand feature relationships

Step 4: Feature Engineering

Important preprocessing steps include:

- Label encoding
- Numeric feature extraction
- Binary attack labeling
- Standardization

Step 5: Dimensionality Reduction

PCA is used to:

- Reduce dimensionality
- Remove redundant information
- Improve clustering efficiency
- Reduce computational complexity

Step 6: Clustering

MiniBatch K-Means groups traffic into clusters based on feature similarity.

Step 7: Anomaly Detection

Two anomaly detection algorithms are applied:

- Isolation Forest
- One-Class SVM

Step 8: Hybrid Analysis

Predictions from multiple models are combined to improve robustness.

Dataset Analysis

Dataset Overview

The dataset contains network flow statistics representing communication between systems.

Dataset Shape

- Rows: ~286,467
- Columns: 79

The features include:

- Packet statistics
- Flow duration
- Packet lengths
- Header information
- Forward/backward traffic statistics
- Timing features
- Traffic behavior indicators

The dataset is highly suitable for anomaly detection because network attacks usually create measurable deviations from normal traffic patterns.

Important Features in the Dataset

Traffic Volume Features

Examples:

- Total Fwd Packets
- Total Backward Packets
- Flow Bytes/s
- Flow Packets/s

These features help identify traffic spikes, DDoS behavior, and abnormal transmission rates.

Timing Features

Examples:

- Flow Duration
- Idle Mean
- Active Mean

Useful for detecting:

- Slow attacks
- Bot behavior
- Persistent scanning

Packet Size Features

Examples:

- Packet Length Mean
- Packet Length Variance

• Avg Packet Size

These features indicate abnormal payload behavior.

Flag Features

Examples:

• SYN Flag Count
• ACK Flag Count
• PSH Flag Count

These are highly important for identifying:

• Port scans
• SYN floods
• Connection abuse

Data Cleaning Analysis

The notebook performs multiple cleaning operations.

1. Whitespace Removal

Extra spaces in column names or string values can create inconsistent processing.

2. Infinite Value Removal

Infinite values generally occur due to division-based metrics such as:

• Bytes per second
• Packets per second

Infinite values can destabilize PCA and distance-based models.

3. Missing Value Handling

Rows with missing values are removed to avoid training instability.

4. Duplicate Removal

Duplicate rows can bias clustering algorithms.

Removing duplicates improves:

• Cluster diversity
• Generalization
• Model fairness

Exploratory Data Analysis (EDA)

EDA Objectives

The EDA phase was designed to:

- Understand class imbalance
- Study feature distributions
- Identify correlations
- Detect noisy features
- Analyze attack concentration

Attack Distribution Analysis

The attack distribution visualization shows that the dataset is highly imbalanced.

This is common in cybersecurity datasets because:

- Normal traffic dominates real-world systems
- Attack traffic is relatively rare

Cybersecurity Insight

This imbalance creates challenges:

- Models may bias toward normal traffic
- Rare attacks become difficult to detect
- Accuracy becomes misleading

Therefore, metrics like:

- Precision
- Recall
- F1 Score

become more important than accuracy.

Correlation Analysis

The heatmap visualizes relationships between features.

Observations

Highly Correlated Features

Certain packet-based statistics are strongly correlated.

Example:

- Flow Bytes/s and Packet Rate
- Packet Length Mean and Avg Packet Size

Redundant Information

Highly correlated features create:

- Multicollinearity
- Redundant dimensions
- Increased computational complexity

This motivates PCA.

Feature Distribution Analysis

Histograms and KDE plots reveal:

- Heavy skewness
- Long-tail distributions
- Outlier concentration
- Non-Gaussian behavior

Cybersecurity Importance

Attack traffic often appears as:

- Extreme outliers
- Dense abnormal clusters
- Sudden spikes

These distributions justify the use of anomaly detection algorithms.

Feature Engineering

Feature engineering is one of the most important parts of the project.

Numeric Feature Extraction

Only numeric columns are used because:

- PCA requires numerical input
- Distance-based models require numeric representation
- Scaling is easier

Binary Label Creation

Although training is unsupervised, binary labels are created for evaluation.

Labels

- 0 → Normal Traffic
- 1 → Attack Traffic

This allows calculation of:

- Accuracy
- Precision
- Recall
- F1 Score

Feature Scaling

StandardScaler is applied.

Why Scaling is Necessary

Without scaling:

- Large-value features dominate clustering
- Distance metrics become biased
- PCA components become distorted

Scaling standardizes features to:

- Mean = 0
- Standard Deviation = 1

This improves:

- PCA performance
- K-Means clustering
- One-Class SVM stability

PCA Analysis

Why PCA Was Used

The original dataset contains 79 features.

High-dimensional data causes:

- Curse of dimensionality
- Sparse distance relationships
- Increased training cost
- Visualization difficulty

PCA solves these issues.

PCA Working Principle

PCA transforms original features into principal components.

Each component:

- Captures maximum variance
- Is orthogonal to others
- Represents compressed information

Explained Variance Analysis

The explained variance plot helps determine:

- How many components preserve information
- Where diminishing returns begin

Interpretation

Early components capture most variance.

This means:

- Many original features are redundant
- Lower-dimensional representation is possible

Benefits of PCA in This Project

1. Faster Computation

Reduced dimensionality lowers training cost.

2. Better Visualization

2D PCA enables cluster plotting.

3. Noise Reduction

Weak or redundant features contribute less.

4. Improved Clustering

Clusters become more separable.

MiniBatch K-Means Deep Dive

Why MiniBatch K-Means?

Traditional K-Means becomes computationally expensive on large datasets.

MiniBatch K-Means improves scalability using:

- Random mini-batches
- Incremental centroid updates
- Faster convergence

This makes it suitable for:

- Large-scale cybersecurity monitoring
- Streaming traffic environments
- Real-time systems

K-Means Working Principle

The algorithm:

1. Initializes cluster centroids
2. Assigns points to nearest centroid
3. Updates centroids iteratively
4. Minimizes within-cluster variance

Number of Clusters

The notebook uses:

- n_clusters = 5

This creates five behavioral traffic groups.

Silhouette Score Analysis

Observed silhouette score:

- ~0.516

Interpretation

A silhouette score above 0.5 indicates:

- Moderate cluster separation
- Reasonably meaningful grouping
- Distinct traffic behavior patterns

This suggests PCA successfully improved cluster structure.

Advantages of MiniBatch K-Means

Speed

Efficient for large datasets.

Scalability

Handles streaming traffic.

Simplicity

Easy to interpret clusters.

Visualization Friendly

Works well with PCA-reduced data.

Weaknesses of MiniBatch K-Means

Sensitive to Initialization

Poor centroid initialization affects results.

Assumes Spherical Clusters

Real attack behavior may not follow spherical shapes.

Difficult with Overlapping Attacks

Some attacks may merge with normal traffic.

Isolation Forest Deep Dive

Core Idea

Isolation Forest isolates anomalies instead of profiling normal points.

Anomalies are:

- Rare
- Different
- Easier to isolate

How Isolation Forest Works

The algorithm:

1. Randomly selects a feature
2. Randomly selects a split value
3. Builds isolation trees
4. Measures path lengths

Shorter path length = higher anomaly probability.

Why Isolation Forest Works Well in Cybersecurity

Network attacks often:

- Produce extreme values
- Deviate strongly from normal traffic
- Create sparse patterns

These properties make attacks easier to isolate.

Isolation Forest Strengths

Efficient on Large Data

Works well with high-volume traffic.

No Distance Calculations

More scalable than distance-based methods.

Good for Outlier Detection

Suitable for rare attack identification.

Isolation Forest Limitations

Struggles with Dense Attacks

Dense attack groups may appear normal.

Sensitive to Contamination Parameter

Incorrect contamination rate affects anomaly detection.

One-Class SVM Deep Dive

Core Idea

One-Class SVM learns the boundary of normal data.

Points outside the boundary are treated as anomalies.

One-Class SVM Working Principle

The algorithm:

1. Maps data into higher-dimensional space
2. Learns a separating boundary
3. Maximizes margin around normal data
4. Flags outside points as anomalies

Why It Is Useful in Cybersecurity

One-Class SVM is effective when:

- Only normal traffic is available
- Attack patterns constantly evolve
- Decision boundaries are nonlinear

Strengths of One-Class SVM

Handles Nonlinear Boundaries

Can capture complex traffic structures.

Good for Unknown Attacks

Suitable for zero-day threat detection.

Weaknesses of One-Class SVM

Computationally Expensive

Training becomes slow on very large datasets.

Sensitive to Hyperparameters

Gamma and nu significantly affect performance.

Scaling Requirement

Strongly dependent on standardized features.

Hybrid Model Analysis

Motivation Behind Hybrid Approach

Each anomaly detection model has different strengths.

Combining them improves:

- Robustness
- Reliability
- Generalization

Hybrid Strategy

The hybrid system combines:

- Clustering-based behavior understanding
- Isolation-based anomaly detection
- Boundary-based anomaly detection

This creates a multi-perspective anomaly detection framework.

Evaluation Metrics Analysis

Model Performance Summary

MiniBatch K-Means

- Accuracy ≈ 0.532
- Precision ≈ 0.063
- Recall ≈ 0.007
- F1 ≈ 0.013

Isolation Forest

- Accuracy ≈ 0.532
- Precision ≈ 0.062
- Recall ≈ 0.007
- F1 ≈ 0.013

One-Class SVM

- Accuracy ≈ 0.531
- Precision ≈ 0.051
- Recall ≈ 0.006
- F1 ≈ 0.011

Hybrid Model

- Accuracy ≈ 0.531
- Precision ≈ 0.061
- Recall ≈ 0.007
- F1 ≈ 0.013

Interpretation of Results

Why Accuracy Looks Misleading

The dataset is highly imbalanced.

Even weak models can achieve high accuracy by predicting mostly normal traffic.

Therefore:

- Precision
- Recall
- F1 Score

are more meaningful.

Low Recall Explanation

Low recall indicates:

- Many attacks remain undetected
- Attack patterns overlap with normal traffic
- Unsupervised learning is challenging for highly complex attack distributions

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Positive Takeaway

Despite limited recall:

- The models successfully identified meaningful abnormal structures
- PCA improved separability

- Clustering revealed behavioral patterns
- Hybrid modeling improved robustness

Detailed Interpretation of All 7 Clustering Visualizations

Visualization 1: Attack Distribution Plot

Purpose

Shows imbalance between normal and attack traffic.

Interpretation

- Normal traffic dominates the dataset.
- Attack traffic is significantly smaller.
- Imbalanced learning conditions exist.

Cybersecurity Insight

Real enterprise traffic behaves similarly.

This confirms the dataset realistically represents production environments.

Visualization 2: Correlation Heatmap

Purpose

Shows relationships between network features.

Interpretation

- Several features are strongly correlated.
- Packet-related features move together.
- Some redundant features exist.

Impact

This justifies PCA-based dimensionality reduction.

Visualization 3: Feature Distribution Plots

Purpose

Understand statistical behavior of features.

Interpretation

- Most features are heavily skewed.
- Extreme outliers are visible.
- Long-tail behavior indicates attack anomalies.

Cybersecurity Meaning

Abnormal traffic often creates:

- Sudden spikes
- Extreme packet counts
- Unusual timing patterns

Visualization 4: PCA Explained Variance Plot

Purpose

Shows how much variance is preserved by principal components.

Interpretation

- Early components capture significant information.
- Later components contribute less.
- Dimensionality can be safely reduced.

Impact

Reduced dimensionality improves clustering quality.

Visualization 5: PCA Cluster Scatter Plot

Purpose

Visualizes clusters in reduced 2D PCA space.

Interpretation

- Five distinct behavioral groups appear.
- Some overlap exists between clusters.
- Dense and sparse traffic groups are visible.

Cybersecurity Meaning

Different traffic behaviors represent:

- Normal browsing
- File transfers
- Scanning activity
- High-volume traffic
- Potential attack regions

Visualization 6: Cluster Attack Ratio Plot

Purpose

Shows percentage of attacks inside each cluster.

Interpretation

Observed attack ratios:

- Cluster 4 shows highest attack concentration
- Other clusters contain mostly benign traffic

Security Meaning

Cluster 4 likely represents:

- Suspicious traffic behavior
- Concentrated anomaly regions
- Potential attack patterns

This is extremely useful for SOC analysts.

Visualization 7: PCA 3D Visualization Graph

Purpose

The 3D PCA visualization projects the high-dimensional network traffic data into three principal components.

This allows:

- Better spatial understanding of cluster separation
- Visualization of anomaly concentration
- Identification of overlapping traffic behavior
- Multi-dimensional behavioral analysis

Interpretation

Distinct Cluster Regions

Several clusters appear visibly separated in 3D space.

This indicates:

- PCA successfully preserved important structural information
- Traffic behaviors contain measurable differences
- Clustering captured meaningful network patterns

Overlapping Regions

Some clusters partially overlap.

This happens because:

- Certain attack traffic resembles normal traffic
- Network behavior patterns can be similar
- Unsupervised learning cannot perfectly separate all attacks

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Dense vs Sparse Regions

The visualization contains:

- Dense central traffic regions
- Sparse outer anomaly regions

Dense regions generally represent:

- Normal enterprise communication
- Frequent network activity
- Stable traffic behavior

Sparse regions usually indicate:

- Rare traffic patterns
- Abnormal communication
- Potential intrusion attempts

Outlier Concentration

A few isolated points appear far from dense traffic clusters.

These outliers are highly important because they may represent:

- DDoS traffic
- Port scanning activity
- Botnet communication
- Malicious packet bursts
- Unknown attack behavior

Cybersecurity Meaning

The 3D PCA visualization is extremely valuable for SOC (Security Operations Center) analysts because it provides visual interpretability of network behavior.

Security teams can:

- Detect suspicious traffic regions
- Identify attack-heavy clusters
- Understand traffic relationships
- Prioritize anomaly investigation
- Monitor evolving attack behavior

Key Insight from the Visualization

The most important observation is that attack-heavy traffic tends to concentrate in specific spatial regions rather than being uniformly distributed.

This confirms that:

- Malicious traffic exhibits distinguishable behavioral characteristics
- PCA successfully compressed meaningful cybersecurity information
- Clustering and anomaly detection can effectively identify suspicious network behavior

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Cluster-by-Cluster Attack Analysis

Cluster 0 Analysis

Attack Ratio

~0.0078

Interpretation

- Mostly benign traffic
- Minor anomalies present
- Likely routine communication

Cluster 1 Analysis

Attack Ratio

~0.0001

Interpretation

- Highly stable normal traffic
- Represents clean communication patterns
- Strong baseline cluster

Cluster 2 Analysis

Attack Ratio

~0.044

Interpretation

- Elevated anomaly concentration
- Suspicious packet behavior
- Potential attack-heavy region

Cluster 3 Analysis

Attack Ratio

~0.0042

Interpretation

- Mostly benign traffic
- Some isolated anomalies
- Medium-risk cluster

Cluster 4 Analysis

Attack Ratio

~0.871

Interpretation

- Extremely attack-dense cluster
- Strong anomaly concentration
- High-risk network behavior region

Security Implication

This cluster is the most critical output of the project.

SOC teams can:

- Prioritize investigation
- Trigger alerts
- Perform packet inspection
- Block suspicious IPs

Cybersecurity Implications

Importance of Unsupervised NIDS

Traditional IDS systems depend on:

- Known attack signatures
- Predefined rules
- Historical attack databases

Unsupervised ML provides:

- Zero-day detection capability
- Adaptive behavior analysis
- Reduced dependency on labels

Practical Security Applications

Enterprise Monitoring

Detect abnormal employee or server behavior.

Cloud Security

Monitor large-scale cloud traffic.

SOC Automation

Assist security analysts using anomaly prioritization.

Threat Hunting

Identify suspicious hidden behaviors.

IoT Security

Detect unusual communication patterns.

Real-World Production Challenges

Data Drift

Traffic behavior changes over time.

Evolving Attacks

Attackers adapt to detection systems.

False Positives

Excessive alerts reduce analyst efficiency.

Scalability

Enterprise networks generate massive traffic volumes.

Why This Project Is Strong for Interviews

This project demonstrates:

- Machine learning fundamentals
- Unsupervised learning knowledge
- Cybersecurity domain understanding
- PCA and dimensionality reduction
- Clustering expertise

• Anomaly detection implementation
• Data preprocessing skills
• Visualization and interpretation ability

It combines both:

• Data science
• Security analytics

which makes it highly valuable.

Limitations

1. Low Recall

Many attacks remain undetected.

2. Imbalanced Dataset

Normal traffic dominates the dataset.

3. Limited Temporal Modeling

Sequential packet relationships are not modeled.

4. No Deep Learning Models

The project does not use:

• Autoencoders
• LSTMs
• Graph neural networks

5. Static Dataset

Real-world streaming data is not used.

6. Hyperparameter Sensitivity

Model performance depends heavily on tuning.

Future Scope

1. Deep Learning-Based NIDS

Future models can use:

- Autoencoders
- Variational Autoencoders
- LSTM Networks
- Transformer architectures

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2. Real-Time Streaming Detection

Integrate:

- Apache Kafka
- Spark Streaming
- Flink

for live anomaly detection.

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3. Explainable AI

Add interpretability using:

- SHAP
- LIME
- Feature attribution methods

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4. Ensemble Learning

Combine:

- Isolation Forest
- Autoencoders
- Clustering
- Graph anomaly detection

for stronger detection.

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5. Federated Learning

Train intrusion models across distributed systems without sharing sensitive data.

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6. Threat Intelligence Integration

Combine anomaly scores with:

- IP reputation
- Threat feeds
- Malware indicators

Interview Questions & Answers

Q1. Why did you choose unsupervised learning for NIDS?

Answer

In cybersecurity, labeled attack data is often incomplete or unavailable. Unsupervised learning helps identify abnormal behavior without requiring predefined attack signatures, making it useful for detecting zero-day attacks.

Q2. Why was PCA necessary?

Answer

The dataset contained 79 features with significant redundancy and correlation. PCA reduced dimensionality, improved clustering efficiency, removed noise, and enabled 2D visualization.

Q3. Why use MiniBatch K-Means instead of normal K-Means?

Answer

MiniBatch K-Means is computationally faster and more scalable because it trains on small batches instead of the full dataset at once.

Q4. What does the silhouette score indicate?

Answer

The silhouette score measures cluster separation quality. A score around 0.516 indicates moderate but meaningful cluster separation.

Q5. Why was StandardScaler used?

Answer

Scaling prevents large-value features from dominating distance calculations and improves PCA, K-Means, and SVM performance.

Q6. What is the difference between Isolation Forest and One-Class SVM?

Answer

Isolation Forest isolates anomalies using random tree splits, while One-Class SVM learns a decision boundary around normal data.

Q7. Why were precision and recall more important than accuracy?

Answer

The dataset is highly imbalanced. Accuracy alone becomes misleading because the model can predict mostly normal traffic and still achieve decent accuracy.

Q8. Why was cluster 4 significant?

Answer

Cluster 4 had the highest attack concentration (~87%), making it the strongest suspicious traffic cluster.

Q9. What are the limitations of unsupervised anomaly detection?

Answer

Unsupervised models may generate false positives and struggle when attack traffic overlaps with normal traffic.

Q10. How would you improve this project?

Answer

I would add deep learning-based anomaly detection, streaming pipelines, explainable AI techniques, and real-time monitoring infrastructure.

Advanced Interview Questions

Q11. Why does PCA improve clustering?

Answer

PCA removes noisy and redundant dimensions, making cluster distances more meaningful.

Q12. What is contamination in Isolation Forest?

Answer

Contamination specifies the expected proportion of anomalies in the dataset.

Q13. Why is One-Class SVM computationally expensive?

Answer

Because it relies on kernel computations and support vectors in high-dimensional space.

Q14. What is the curse of dimensionality?

Answer

In high-dimensional space, distance relationships become less meaningful, making clustering and anomaly detection more difficult.

Q15. How would you deploy this project?

Answer

I would containerize the pipeline using Docker, expose inference APIs using FastAPI, and deploy monitoring dashboards with Kafka-based streaming integration.

Resume Points

Resume Point 1

Developed a machine learning-based Network Intrusion Detection System using unsupervised learning algorithms including MiniBatch K-Means, Isolation Forest, and One-Class SVM.

Resume Point 2

Performed extensive preprocessing and feature engineering on a cybersecurity dataset containing 286K+ network traffic records and 79 features.

Resume Point 3

Applied Principal Component Analysis (PCA) for dimensionality reduction and visualization of network behavior patterns.

Resume Point 4

Built clustering-based anomaly detection pipelines capable of identifying suspicious network traffic patterns and high-risk attack clusters.

Resume Point 5

Implemented hybrid anomaly detection strategies and evaluated models using accuracy, precision, recall, F1-score, and silhouette score.

Resume Point 6

Created visual analytics dashboards and cluster interpretation workflows for cybersecurity behavior analysis.

Technical Skills Demonstrated

Machine Learning

- PCA
- Clustering
- Anomaly Detection
- Unsupervised Learning

Python Libraries

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Cybersecurity Skills

- Intrusion Detection
- Network Traffic Analysis
- Attack Pattern Analysis
- Threat Detection

Data Engineering Skills

- Data Cleaning
- Feature Scaling
- Dimensionality Reduction
- Visualization

Key Learning Outcomes

This project teaches:

- End-to-end ML pipeline development
- Real-world anomaly detection challenges
- Cybersecurity data interpretation
- Unsupervised learning techniques
- Visualization-driven analysis
- Model evaluation in imbalanced datasets

Final Conclusion

This project demonstrates the practical application of unsupervised machine learning for cybersecurity intrusion detection.

The workflow combines:

- Data preprocessing
- Feature engineering
- PCA dimensionality reduction
- Clustering
- Anomaly detection
- Hybrid modeling

to identify suspicious network behavior.

MiniBatch K-Means successfully revealed behavioral traffic clusters, while Isolation Forest and One-Class SVM detected anomalous patterns using different mathematical approaches.

The project highlights both the strengths and limitations of unsupervised anomaly detection in cybersecurity environments.

Key insights include:

- PCA significantly improved cluster interpretability
- Cluster 4 represented a highly suspicious attack-heavy region
- Hybrid modeling improved robustness
- Imbalanced cybersecurity datasets require careful metric interpretation

Although recall values remain low, the project provides a strong foundation for:

- Real-time intrusion detection
- Threat hunting systems
- Security analytics pipelines
- AI-driven cybersecurity monitoring

From an interview perspective, this project strongly demonstrates:

- ML fundamentals
- Cybersecurity understanding
- Data preprocessing expertise
- Visualization and interpretation skills
- Practical anomaly detection implementation

It is a highly valuable portfolio project for roles such as:

- Machine Learning Engineer
- Data Scientist
- Cybersecurity Analyst

- Security Data Analyst
- AI Security Researcher

Quick Interview Revision Cheat Sheet

Problem Statement

Build an unsupervised Network Intrusion Detection System capable of identifying suspicious network traffic.

Algorithms Used

- PCA
- MiniBatch K-Means
- Isolation Forest
- One-Class SVM
- Hybrid Ensemble

Key Metrics

- Silhouette Score ≈ 0.516
- Best attack cluster ratio ≈ 87%

Major Challenges

- High dimensionality
- Imbalanced dataset
- Rare attacks
- False positives

Major Learnings

- PCA improves clustering
- Scaling is critical
- Unsupervised learning is difficult but realistic for cybersecurity
- Visualization helps interpret anomalies

Best Talking Point

"The most important insight from the project was discovering that one cluster contained nearly 87% attack concentration, showing how clustering can reveal hidden malicious behavior patterns in network traffic."